

Confidence Intervals In Multiple Regression and Model Specification

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April 17, 2017

Chapter Outline

- Confidence sets for multiple coefficients.
- Model specification for multiple regression.

Confidence Sets for Multiple Coefficients

Recall the meaning of a 95% confidence interval

- The set of points that cannot be rejected at the 5% significance level;
- a set-valued function of the data (an interval) containing the true parameter value 95% of the time in hypothetical repeated samples.
- Example, if our regression output is

$$\widehat{TestScore} = \underset{(10.4)}{698.9} - \underset{(0.52)}{2.28} \times STR,$$

$$R^2 = 0.05, \quad SER = 18.6.$$

the 95% confidence interval for β_1 is $[-3.30, -1.26]$.

- If we tested the null hypothesis $H_0 : \beta_1 = \beta_{1,0}$ vs. $\beta_1 \neq \beta_{1,0}$ using a *t-statistic* at the 5% significance level, we would not reject H_0 for all the values of $\beta_{1,0}$ between -3.3 and -1.25 .

Confidence Sets for Multiple Coefficients

Recall our multiple regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_K X_{Ki} + u_i, \quad \forall i = 1, \dots, n. \quad (1)$$

What is a joint confidence set for β_1 and β_2 ?

- A 95% joint confidence set is: the set of coefficient values that cannot be rejected at the 5% significance level.
- Equivalently, a set-valued function of the data that contains the true coefficient(s) in 95% of hypothetical repeated samples.

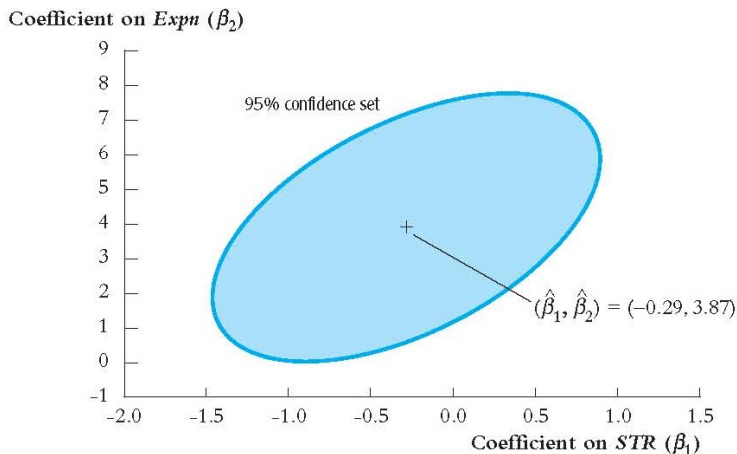
Confidence Sets for Multiple Coefficients

- Imagine you want to construct a 95% confidence set for β_1 and β_2 .
- We learned before how to use the *F-statistic* to test the joint null hypothesis that $H_0 : \beta_1 = \beta_{1,0}$ and $\beta_2 = \beta_{2,0}$ at the 95% confidence level.
- Suppose you were able to test all the possible values of $\beta_{1,0}$ and $\beta_{2,0}$ and save all the pairs $(\beta_{1,0}, \beta_{2,0})$ for which the F-statistic is less than the critical value 3. The set of values above represent the 95% confidence set.
- The procedure described above is good to build intuition but it is impractical because you would need to test for potentially an extremely large number of combinations of values $(\beta_{1,0}, \beta_{2,0})$.

Confidence Sets for Multiple Coefficients

- You can find a 95% confidence set as the set of $(\beta_{1,0}, \beta_{2,0})$ that cannot be rejected at the 5% level using an F-test.
- Let $F(\beta_{1,0}, \beta_{2,0})$ be the (heteroskedasticity-robust) *F-statistic* testing the hypothesis $H_0 : \beta_1 = \beta_{1,0}$ and $\beta_2 = \beta_{2,0}$. The 95% confidence set is $\{(\beta_{1,0}, \beta_{2,0}) : F(\beta_{1,0}, \beta_{2,0}) \leq 3.00\}$.
- 3.00 is the 5% critical value of the $F_{2,\infty}$ distribution.
- This set has coverage rate 95% because the test on which it is based (i.e., the test it “inverts”) has a size of 5%.
- 5% of the time, the test incorrectly rejects the null when the null is true, so 95% of the time it does not; therefore the confidence set constructed as the non-rejected values contains the true values 95% of the time (in 95% of all hypothetical samples).

Confidence Sets for Multiple Coefficients



Model Specification for Multiple Regression

- We want to get an unbiased estimate of the effect on test scores of changing class size, holding constant factors outside the school committee's control –such as outside learning opportunities (museums, etc), parental involvement in education (reading with mom at home?), etc.
- If we could run an experiment, we would randomly assign students (and teachers) to different sized classes. Then STR_i would be independent of all the things that go into u_i and $E[u_i|STR_i] = 0$ and the *OLS* slope estimator in the regression of $TestScore_i$ on STR_i will be an unbiased estimator of the desired causal effect.

Model Specification for Multiple Regression

- But with observational data, u_i depends on additional factors (museums, parental involvement, knowledge of English, etc).
- If you can observe those factors (e.g. $PctEL$), then include them in the regression.
- But usually you can't observe all these omitted causal factors (e.g. parental involvement in homework).
- If additional relevant variables are not observable, sometimes you can include “control variables” which are correlated with these omitted causal factors, but which themselves are not causal.

Model Specification for Multiple Regression

Control Variables

- A control variable W is a variable that is correlated with, and controls for, an omitted causal factor in the regression of Y on X , but which itself does not necessarily have a causal effect on Y .
- Control variables are **NOT** the object of interest; rather they are regressors we include to hold constant factors that, if neglected, could lead to the problem of omitted variable bias.

Model Specification for Multiple Regression

Consider the following regression

$$\widehat{TestScore} = 700.2 - \underset{(5.6)}{1.00} \times STR, - \underset{(0.033)}{0.122} \times PctEL - \underset{(0.024)}{0.547} \times LchPct, \quad (2)$$

$$\overline{R}^2 = 0.773,$$

where $PctEL$ is the percentage of English learners in the school district, and $LchPct$ is the percentage of students receiving a free/subsidized lunch (only students from low-income families are eligible).

Which variable is the variable of interest? Which variables are control variables? Do they have causal components? What do they control for?

Model Specification for Multiple Regression

Answers:

- *STR is the variable of interest.*
- *PctEL* probably has a direct causal effect (school is tougher if you are learning English!). But it is also a control variable: immigrant communities tend to be less affluent and often have fewer outside learning opportunities, and *PctEL* is correlated with those omitted causal variables. *PctEL is both a possible causal variable and a control variable.*
- *LchPct* might have a causal effect (eating lunch helps learning); it also is correlated with and controls for income-related outside learning opportunities. *LchPct is both a possible causal variable and a control variable.*

Model Specification for Multiple Regression

Three interchangeable statements about what makes an effective control variable:

- 1 An effective control variable is one which, when included in the regression, makes the error term uncorrelated with the variable of interest.
- 2 Holding constant the control variable(s), the variable of interest is “as if” randomly assigned.
- 3 Among individuals (entities) with the same value of the control variable(s), the variable of interest is uncorrelated with the omitted determinants of Y .

Model Specification for Multiple Regression

Control variables need not be causal, and their coefficients generally do not have a causal interpretation.

$$\widehat{TestScore} = 700.2 - \underset{(5.6)}{1.00} \times STR, -\underset{(0.033)}{0.122} \times PctEL - \underset{(0.024)}{0.547} \times LchPct,$$

Does the coefficient on $LchPct$ have a causal interpretation?

If so, then we should be able to boost test scores (by a lot!) by simply eliminating the school lunch program so that $LchPct = 0$!

Model Specification for Multiple Regression

Conditional Mean Independence

When control variables are included, the *OLS* (1) $E[u_i | X_{1,i}, \dots, X_{K,i}] = 0$ must not hold.

For example, the coefficient on *LchPct* is correlated with unmeasured determinants of test scores such as outside learning opportunities. But the fact that *LchPct* is correlated with these omitted variables is precisely what makes it a good control variable!

Model Specification for Multiple Regression

Conditional Mean Independence

- If *OLS (1)* does not hold, then what does?
- We need a mathematical statement of what makes an effective control variable. This condition is called **conditional mean independence**: given the control variable, the mean of u_i does not depend on the variable of interest.
- Let X_i denote the variable of interest and W_i denote the control variable(s). W is an effective control variable if conditional mean independence holds: $E[u_i|X_i, W_i] = E[u_i|W_i]$
- If W is a control variable, then conditional mean independence replaces *OLS (1)* –it is the version of *OLS (1)* which is relevant for control variables.

Model Specification for Multiple Regression

Conditional Mean Independence

$$Y = \beta_0 + \beta_1 X + \beta_2 W + u, \quad (3)$$

where X is the variable of interest and W is an effective control variable so that conditional mean independence holds if $E[u_i|X_i, W_i] = E[u_i|W_i]$.

Model Specification for Multiple Regression

Regression With Control Variables

Suppose that in addition to conditional mean independence the Least Square Assumptions (2) and (3) hold, then

- β_1 has a causal interpretation.
- $\hat{\beta}_1$ is unbiased.
- The estimator of the coefficient on the control variable, $\hat{\beta}_2$ is, in general, biased. This is because the control variable is correlated with omitted variables in the error term, so that is subject to omitted variable bias.
- This is not a problem for us because we do not aim to interpret the coefficient of the control variable.

Model Specification in Theory and in Practice

- 1. Identify the variable/s of interest.
- 2. Think of the omitted causal effects that could result in omitted variable bias.
- 3. Include those omitted causal effects if you can or, if you can not, include variables correlated with them that serve as control variables. The control variables are effective if the conditional mean independence assumption plausibly holds (i.e., if u is uncorrelated with the variable/s of interest once the control variables are included). This results in a “base” or “benchmark” model.
- 4. Specify a range of plausible alternative models, which include additional candidate variables.

Model Specification in Theory and in Practice

- 5. Estimate your base model and plausible alternative specifications (“sensitivity checks”).
 - Does a candidate variable change the coefficient of interest (β 's)?
 - Is a candidate variable statistically significant?
 - Use judgement, not a mechanical recipe...
- 6. Do not just try to maximize R^2 ! It is easy to fall into the trap of maximizing the R^2 and $\bar{R}^2$, but this loses sight of our real objective; an unbiased estimator of the parameter of interest. Remember, R^2 and $\bar{R}^2$ are measures of fit, they just tell you whether regressors are good at predicting the values of the dependent variable in the sample, not whether there exist a causal relationship between dependent and independent variables.

Interpreting the R^2 and $\overline{R}^2$

- An increase in R^2 or $\overline{R}^2$ does not necessarily mean that an added variable is statistically significant –this must be determined using hypotheses tests.
- A high R^2 or $\overline{R}^2$ does not mean the regressors are a true cause of the dependent variable.
- A high R^2 or $\overline{R}^2$ does not mean you have eliminated omitted variable bias.
- A high R^2 or $\overline{R}^2$ does not mean that you have the most appropriate set of regressors, nor does a low R^2 or $\overline{R}^2$ necessarily mean that you have an inappropriate set of regressors.

Analysis of the Test Score Data Set

- 1. Identify the variable/s of interest.

Test Scores and STR

- 2. Possible omitted variables.

Whether the students know English; outside learning opportunities; parental involvement; teacher quality (if teacher salary is correlated with district wealth) –there is a long list!

- 3. Include those omitted causal effects if you can or, if you can not, include control variables.

Many of the omitted causal variables are hard to measure, so we need to find control variables. These include PctEL (both a control variable and an omitted causal factor) and measures of district wealth.

Analysis of the Test Score Data Set

- 4. Specify a range of plausible alternative models, which include additional candidate variables.

It is not clear which of the income-related variables will best control for the many omitted causal factors such as outside learning opportunities, so the alternative specifications include regressions with different income variables. The alternative specifications considered here are just a starting point, not the final word!

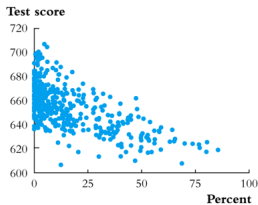
Analysis of the Test Score Data Set

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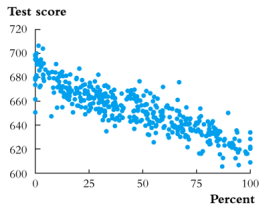
We consider three variables controlling for background characteristics of the students that could affect test scores: the fraction of students who are still learning English, the percentage of students who are eligible for receiving a subsidized or free lunch at school, and the percentage of students in the district whose families qualify for California income assistance program

Case Study

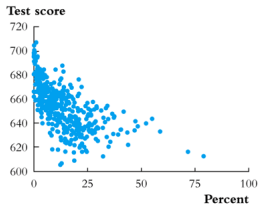
FIGURE 7.2 Scatterplots of Test Scores vs. Three Student Characteristics



(a) Percentage of English language learners



(b) Percentage qualifying for reduced price lunch



(c) Percentage qualifying for income assistance

A Digression on Reporting Results

- Different model specifications lead to a number of regressions that we want to report. It is awkward and difficult to read several regressions written out in equation form (as we did in chapter 5). Instead it is conventional to report them in a table.
- A table of regression results should include
 - Estimated regression coefficients.
 - Standard errors.
 - Measures of fit.
 - Number of observations.
 - Relevant F-statistics, if any.
 - Any other pertinent information.

Case Study

TABLE 7.1 Results of Regressions of Test Scores on the Student–Teacher Ratio and Student Characteristic Control Variables Using California Elementary School Districts

Dependent variable: average test score in the district.

Regressor	(1)	(2)	(3)	(4)	(5)
Student–teacher ratio (X_1)	−2.28** (0.52)	−1.10* (0.43)	−1.00** (0.27)	−1.31** (0.34)	−1.01** (0.27)
Percent English learners (X_2)		−0.650** (0.031)	−0.122** (0.033)	−0.488** (0.030)	−0.130** (0.036)
Percent eligible for subsidized lunch (X_3)			−0.547** (0.024)		−0.529** (0.038)
Percent on public income assistance (X_4)				−0.790** (0.068)	0.048 (0.059)
Intercept	698.9** (10.4)	686.0** (8.7)	700.2** (5.6)	698.0** (6.9)	700.4** (5.5)
Summary Statistics					
<i>SER</i>	18.58	14.46	9.08	11.65	9.08
$\bar{R}^2$	0.049	0.424	0.773	0.626	0.773
<i>n</i>	420	420	420	420	420

These regressions were estimated using the data on K-8 school districts in California, described in Appendix 4.1. Heteroskedasticity-robust standard errors are given in parentheses under coefficients. The individual coefficient is statistically significant at the *5% level or **1% significance level using a two-sided test.